

Peripherals & Connectivity

1. Connect five different types of peripherals (one each from input, output, storage, and communication categories) to your PC and document the connection method (USB, Bluetooth, Ethernet, etc.), including any visible device name in Device Manager or System Preferences.

Hint: Example peripherals: USB mouse, external hard drive, headphones, network adapter, webcam.

->Here is an **assignment-ready example table**. Replace the device names with the exact names shown on your PC.

No.	Peripheral	Category	Connection Method	Device Manager / System Name
1	USB Mouse	Input	USB	HID-compliant mouse
2	USB Webcam	Input	USB	USB Camera / HD Webcam
3	Headphones	Output	3.5 mm audio jack / USB	Headphones / USB Audio Device
4	External Hard Drive	Storage	USB 3.0	USB Mass Storage Device
5	Network Adapter	Communication	Ethernet	Realtek PCIe GbE Family Controller

Documentation Steps

1. Connect each peripheral to the appropriate port.
2. Wait for Windows to detect and install the device.
3. Open **Device Manager** by right-clicking the Start button → **Device Manager**.
4. Find the corresponding category, such as **Mice and other pointing devices**, **Cameras**, **Sound, video and game controllers**, **Disk drives**, or **Network adapters**.
5. Record the exact device name displayed.

6. Take a screenshot of Device Manager and photographs of the connected peripherals for your assignment.

Short Observation

All five peripherals were successfully connected using different connection methods. Windows automatically detected the devices and displayed their names in Device Manager, confirming that the peripherals were recognized correctly.

2. Install drivers for a USB device manually by downloading the correct driver from the manufacturer's website, and then verify the device is working correctly in Device Manager (Windows) or System Information (Mac).

->Manual USB Driver Installation – Windows

You can use a **USB Wi-Fi adapter, USB printer, webcam, or other USB device** for this practical.

Step-by-Step Procedure

1. Identify the USB device

- Connect the USB device to the PC.
- Note its **manufacturer and model number**.

2. Download the correct driver

- Visit the manufacturer's **official website**.
- Search for the exact model.
- Select your Windows version, such as Windows 10/11 64-bit.
- Download the appropriate driver package.

3. Install the driver

- Open the downloaded driver file.
- Follow the installation instructions.
- Restart the computer if requested.

4. Check Device Manager

- Right-click **Start → Device Manager**.

- Locate the device under its appropriate category.
- Right-click the device → **Properties** → **Driver**.
- Check the driver provider, version, and date.

5. Verify operation

- Confirm that there is **no yellow warning icon** beside the device.
- In Properties → **General**, the status should indicate that the device is working properly.
- Test the device by using its normal function.

Example Documentation

Item	Result
USB device	USB Wi-Fi Adapter
Connection	USB
Driver source	Manufacturer's official website
Installation	Completed successfully
Device Manager name	USB Wireless Adapter
Warning icon	None
Device status	This device is working properly
Final result	Device working correctly

Tip: Always download drivers from the device manufacturer's official support page rather than random driver-download websites. Use the **exact model and Windows version** to avoid installing an incompatible driver.

3. Set up a network printer (not USB) using its IP address on your local Wi-Fi or LAN, install the correct printer driver, and print a test page to confirm connectivity.

->**Set Up a Network Printer Using Its IP Address – Windows**

Step-by-Step Procedure

1. Connect the printer to the network

- Connect the printer to the same **Wi-Fi or LAN network** as your PC.
- Open the printer's network settings and note its **IP address** (for example, 192.168.1.50).

2. Add the printer using its IP address

- Go to **Settings → Bluetooth & devices → Printers & scanners**.
- Select **Add device**.
- Choose **Add manually** when the printer is not automatically detected.
- Select **Add a printer using an IP address or hostname**.
- Enter the printer's IP address.
- Select the appropriate device type and continue.

3. Install the printer driver

- Download the correct driver for the exact printer model from the **manufacturer's official website**.
- Install the driver when Windows asks for it.
- Select the correct printer model during installation.

4. Verify the printer

- Open **Settings → Printers & scanners** and select the newly added printer.
- Check that it shows as available/ready.
- Open **Printer properties** and verify the IP address and driver information.

5. Print a test page

- Open **Printer properties → General → Print Test Page**.
- Wait for the test page to print.
- Confirm that the page prints correctly.

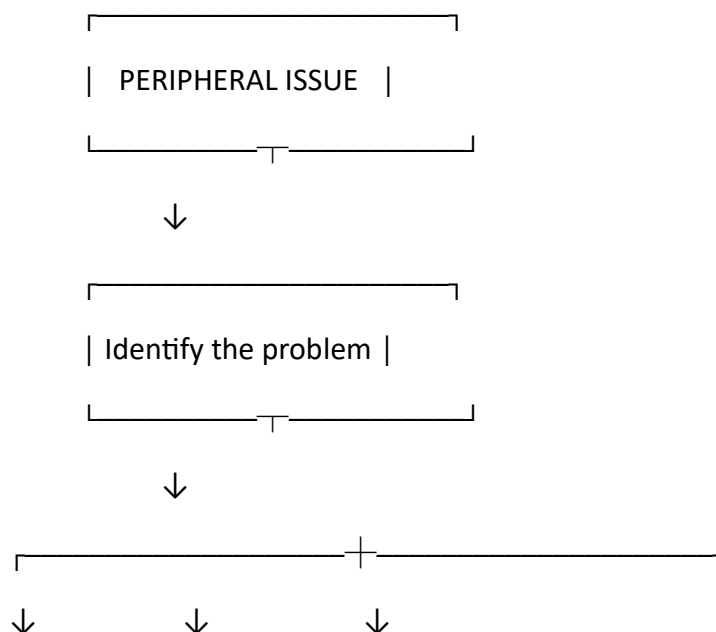
Practical Record

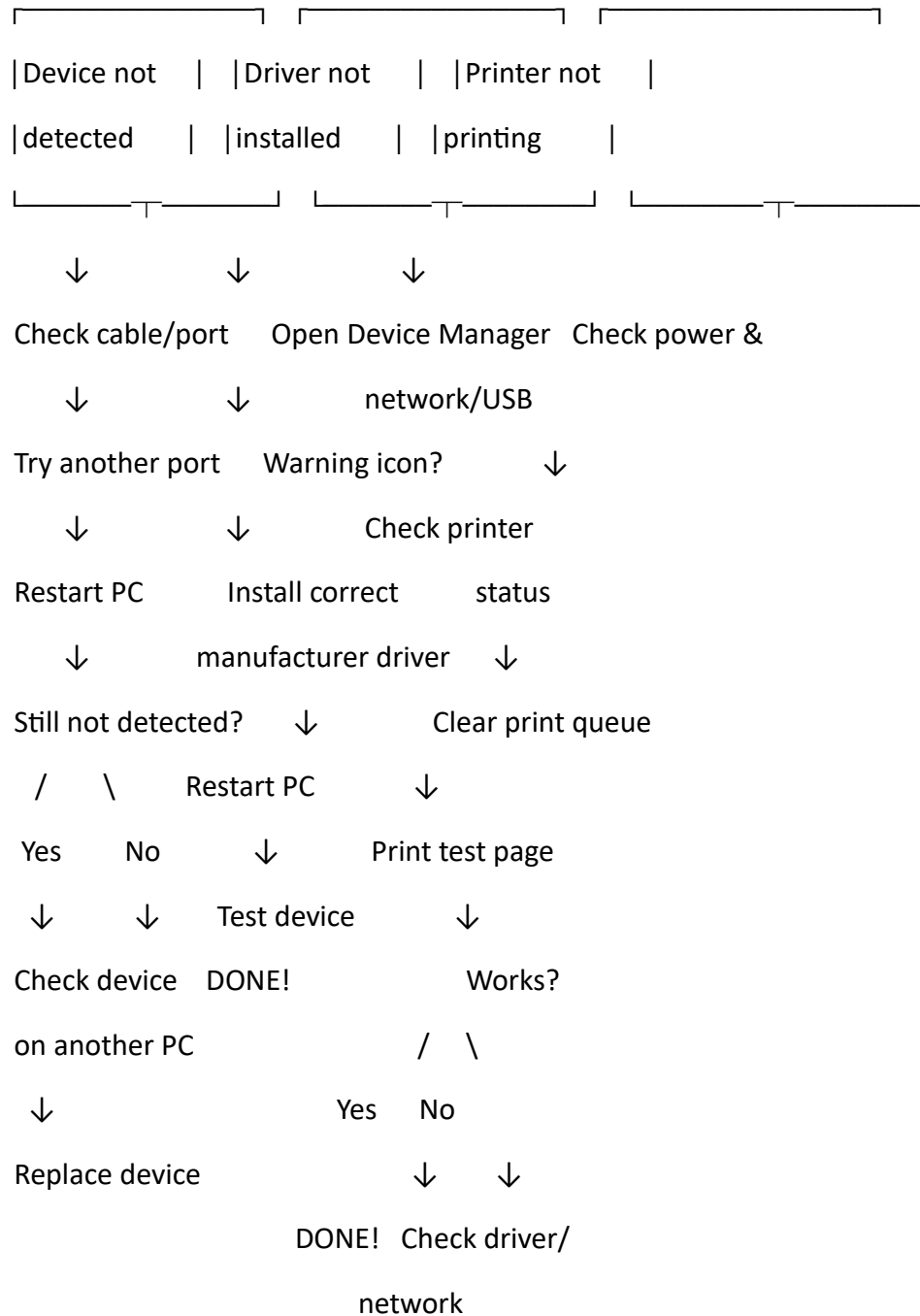
Item	Result
Printer connection	Wi-Fi/LAN
Printer IP address	192.168.1.50 (<i>example</i>)
Printer model	Your actual model
Driver	Correct manufacturer driver
Connection status	Ready/Online
Test page	Printed successfully
Final result	Network printer working correctly

Note: The IP address above is only an example. Record the **actual IP address shown on your printer**.

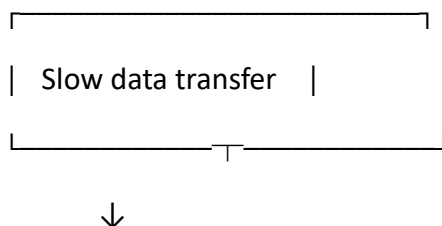
4. Create a troubleshooting flowchart (hand-drawn or digital) that covers common peripheral issues such as 'Device not detected', 'Driver not installed', 'Printer not printing', and 'Slow data transfer', showing the steps to diagnose and solve each problem.

->Here is a clear **assignment-ready troubleshooting flowchart** you can recreate digitally or draw by hand.





Slow Data Transfer



Check USB/network cable



Try another port



Check USB 2.0 vs USB 3.x



Check drive free space/health



Update device/chipset drivers



Test transfer again



Speed improved?

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Yes No



DONE! Check drive,

cable, or

network hardware

Common Troubleshooting Rules

- **Device not detected:** Check connections, try another port, restart the PC, and test the device on another computer.
- **Driver not installed:** Check Device Manager and install the correct driver from the manufacturer's official website.
- **Printer not printing:** Check power, network connection, printer status, paper/ink, print queue, and driver.

- **Slow data transfer:** Check the cable and port, verify the connection standard, check drive health/free space, and update drivers.

5. Design a printer management plan for a startup office of 50 employees: specify how you would organize network printers, manage driver installations, and ensure everyone has access without printer queue conflicts.

Hint: Think about printer sharing, access permissions, and using printer queues or print servers.

->Printer Management Plan for a 50-Employee Startup Office

1. Printer Setup

For 50 employees, I would use **3–4 network printers** rather than connecting printers directly to individual PCs.

Printer	Location	Main Use	Suggested Access
Printer 1	Reception/Admin	General documents	All employees
Printer 2	Finance/HR	Confidential documents	Finance & HR only
Printer 3	Development/Operations	Daily office printing	Development & Operations
Printer 4	Meeting/Management area	Presentations/reports	Management + selected users

All printers should connect directly to the **office LAN/Wi-Fi**, preferably using fixed/reserved IP addresses.

2. Centralized Print Server

Use a **Windows print server** or another centralized print-management system.

PC → Print Queue → Print Server → Network Printer

This gives the administrator centralized control over printer drivers, permissions, and print jobs.

3. Driver Management

- Install the correct manufacturer driver on the print server.
- Use standardized drivers for each printer model.
- Deploy printers/drivers automatically to employees' computers where possible.
- Keep drivers updated and test updates before deploying them to everyone.

- Avoid having employees download random printer drivers themselves.

4. Access Permissions

Create groups such as:

- **All Employees:** General printer
- **Finance/HR:** Confidential printer
- **Management:** Management printer
- **IT Administrators:** All printers and management settings

Users should only see or print to the printers they are authorized to use.

5. Preventing Queue Conflicts

Each printer should have its **own dedicated print queue** with a clear name, for example:

- Office-General
- Finance-HR-Confidential
- Dev-Operations
- Management

The print server processes jobs in the queue so multiple employees can submit documents without sending conflicting jobs directly to the printer.

6. Maintenance and Monitoring

The IT administrator should:

- Monitor printer status and print queues.
- Remove stuck or failed print jobs.
- Check toner/ink and paper levels.
- Keep printer firmware and drivers updated.
- Review printer usage monthly.
- Maintain a backup printer or contingency plan for critical departments.

Overall Plan

Centralized network printers + print server + standardized drivers + user permissions + separate queues provide a scalable setup for 50 employees. This reduces duplicate printer installations, prevents unauthorized printing, and makes troubleshooting much easier.